Fair Decoder Baselines and Rigorous Finite-Size Scaling for Bivariate Bicycle Codes on the Quantum Erasure Channel

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Abstract

We establish a methodological standard for fair threshold estimation of bivariate bicycle (BB) codes on the quantum erasure channel, addressing two systematic issues common in the QLDPC simulation literature: decoder-baseline unfairness and the conflation of finite-size pseudo-thresholds with true asymptotic thresholds. We implement both uninformed and *erasure-aware* minimum-weight perfect matching (MWPM) surface code baselines and show that, under standard depolarizing-weight MWPM without erasure information, performance empirically matches random guessing on the erasure channel within our tested regime, confirming that prior comparisons using this baseline measure decoder quality rather than code quality. Under the fair erasure-aware comparison, a Monte Carlo framework with belief propagation and ordered statistics decoding (BP-OSD) is applied to five BB code sizes from $N = 144$ to $N = 1296$ physical qubits at 200,000 shots per point with rigorous bootstrap confidence intervals. Finite-size pseudo-thresholds (WER = 0.10 crossings) scale from $p^* = 0.370$ to 0.471; finite-size scaling (FSS) analysis extrapolates an asymptotic threshold of $p^*_\infty \approx 0.488$—within 2.4% of the zero-rate limit, achieved by BP-OSD without maximum-likelihood decoding. Under the fair baseline, BB codes at $N = 1296$ achieve a modest threshold advantage ($\Delta p^* \approx 0.011$) over surface codes at twice the physical qubit count, while requiring 12× fewer physical qubits per logical qubit—a reduction in normalized overhead that constitutes the primary practical case for BB codes within this comparison setting. All simulations are fully reproducible with recorded random seeds and package versions.

1 Introduction

Quantum error correction (QEC) is essential for fault-tolerant quantum computation [2, 3]. Quantum low-density parity-check (QLDPC) codes have emerged as promising candidates due to their high error thresholds and moderate encoding rates [1, 4, 5]. Bivariate bicycle codes, introduced by Bravyi et al. [1], represent a particularly attractive family: they achieve large distance with constant overhead and have demonstrated strong threshold properties on erasure channels.

The erasure channel is a natural benchmark for quantum codes because error *locations* are known (e.g., via measurement flagging [8]), allowing erasure-informed decoders to achieve near-optimal performance. A persistent challenge in comparing code families is that unfair decoder choices can artificially inflate apparent advantages. In this work we address this directly by implementing and reporting both an uninformed and an erasure-aware MWPM surface code baseline alongside BP-OSD decoding of BB codes.

We also address a common conflation in the literature between *finite-size pseudo-thresholds* (WER-

crossing points at a fixed target WER) and true asymptotic thresholds. We perform finite-size scaling (FSS) analysis to extract the latter rigorously. Our extrapolated threshold $p^*_\infty \approx 0.488$ approaches within 2.4% of the zero-rate limit under BP-OSD decoding, without requiring maximum-likelihood decoding.

Our main contributions are:

- A reproducible Monte Carlo framework with full X+Z sector decoding and 200,000 shots per point, yielding bootstrap 95% CIs of $\lesssim 0.001$ on pseudo-thresholds. Full reproducibility is enforced via seed tracking, per-point worker seeds, pinned package versions, and JSON metadata files—a level of rigor uncommon in the QLDPC simulation literature.
- A systematic FSS analysis of BB codes on the erasure channel, giving $p^*_\infty \approx 0.488$ (statistical CI ± 0.001; estimated systematic uncertainty $\lesssim 0.005$) and $\nu \approx 1.18$, with explicit treatment of corrections-to-scaling and fit-window sensitivity.
- A fair comparison to erasure-aware MWPM surface codes showing a 12× reduction in normalized overhead for BB codes at comparable threshold, within this comparison setting.
- An explicit erasure-aware fairness correction showing that, under standard depolarizing-weight MWPM without erasure information, surface code baselines empirically match random guessing within our tested regime—a result that should disqualify this baseline as a primary comparison metric in erasure-channel studies.

Throughout this work we emphasize that comparisons between BB and surface codes are not architecture-neutral: BB codes are non-local and require long-range qubit connectivity, whereas surface codes are geometrically local and implementable with nearest-neighbor interactions. The overhead and threshold comparisons should therefore be interpreted as measures of *code efficiency* within the code-capacity setting, not as direct hardware-level claims.

Several recent works study BB codes under erasure or erasure-biased noise, including Pecorari and Pupillo [20], Bhave et al. [21], and Berthusen et al. [19]. Improved decoders for QLDPC codes on related noise models have been investigated by Hillmann et al. [22], Liang et al. [23], Ott et al. [24], Yin et al. [25], and Gong et al. [26]; limitations of BP-based approaches are discussed in iOlius et al. [28]. Additional BB code constructions and families are analyzed by Rabeti and Mahdavifar [27]. Relative to this body of work, our study is distinguished by: (i) an explicit erasure-aware fairness correction that resolves a systematic baseline error common in prior comparisons; (ii) full bootstrap statistical uncertainty at 200,000 shots per point; (iii) FSS with separate statistical and systematic uncertainty budgets; and (iv) exact reproducibility via seed and version tracking.

2 Methods

2.1 Code Construction

Bivariate bicycle codes are constructed using the lifted product of cyclic matrices over $\mathbb{Z}_L \times \mathbb{Z}_M$. For parameters (L, M) the code has $N = 2LM$ physical qubits. For the specific family studied here the generator polynomials are:

$$A = x^3 + y + y^2, \qquad B = y^3 + x + x^2, \tag{1}$$

with parity check matrices $H_x = [A \mid B]$ and $H_z = [B^T \mid A^T]$. The $[[144, 12, 12]]$ Gross code corresponds to $(L, M) = (12, 6)$. This family was chosen because it is the primary BB code studied in Bravyi et al. [1] and subsequent literature, making it the natural benchmark for comparison; results may differ for other polynomial pairs (see Sec. 4.4).

2.2 Simulation Framework

Each Monte Carlo trial proceeds as:

1. **Erasure**: each qubit erased independently with probability p.
2. **Error**: erased qubits receive independent Pauli X and Z errors with Pr = 0.5.

3. **Decoding**: BP-OSD with erasure-informed priors (erased qubits: channel probability 0.5; non-erased: 10^{-10}).

4. **Validation**: logical error if either X or Z sector fails.

WER is the fraction of trials with a logical error in either sector.

2.3 Decoder Configuration

We use the BP-OSD decoder from the `bposd` package [9]: min-sum BP, OSD-CS post-processing, OSD order 10, maximum 50 BP iterations.

2.4 Pseudo-Threshold Estimation

We measure the *pseudo-threshold* p^*: the erasure rate at which WER $= 0.10$ for a given finite code size. This is a size-dependent quantity; true asymptotic thresholds require FSS extrapolation (Section 2.5).

Pseudo-thresholds are located by an adaptive search:

1. Step upward from $p = 0.38$ in increments of 0.04 until WER > 0.10 (bracketing).

2. Refine the bracket using regula falsi (false-position) interpolation with the Illinois modification to prevent stagnation.

3. Stop when bracket width $< 5 \times 10^{-4}$ or after at most 10 total evaluations.

4. Report p^* as the final linear interpolant within the tight bracket.

Each evaluation uses 200,000 shots. Uncertainty is quantified by 5,000-iteration parametric bootstrap resampling from the binomial distribution at the bracketing points, yielding 95% CIs typically $\lesssim 0.001$.

2.5 Finite-Size Scaling

Near the asymptotic threshold p^*_∞, standard FSS theory predicts

$$\mathrm{WER}(p, N) \approx f\left[(p - p^*_\infty)\, N^{1/\nu}\right], \tag{2}$$

where ν is the correlation-length critical exponent and f is a universal scaling function. We approximate f by a degree-3 polynomial and minimize the residual sum of squares over (p^*_∞, ν) using the per-size WER data within a window $|p - p^*| < 0.06$. Bootstrap CIs are obtained from 500 resampled datasets. We do not claim universality of the polynomial ansatz; rather, it serves as a low-order phenomenological model consistent with standard FSS practice, and its adequacy is assessed through fit-window sensitivity and the independent linearized check described below.

This leading-order FSS ansatz assumes that corrections-to-scaling are negligible at the system sizes studied ($N \leq 1296$), which is an uncontrolled approximation. To assess sensitivity, we repeat the fit with windows $|p - p^*| < 0.04$ and $|p - p^*| < 0.08$ and find that p^*_∞ shifts by $\lesssim 0.004$, consistent with our stated systematic uncertainty of $\lesssim 0.005$. An independent check using the linearized FSS relation $p^*(N) \approx p^*_\infty + c\, N^{-1/\nu}$ is reported alongside the primary fit.

2.6 Baseline Comparison

We compare BB codes against two surface code (toric code) baselines, both decoded with MWPM via PyMatching [10]:

Uninformed MWPM. The erasure rate p is converted to an effective depolarizing rate $q = p/2$ and MWPM runs with constant edge weights. This is the most conservative baseline and provides a lower bound on surface code performance.

Erasure-aware MWPM. Erased qubits are assigned weight 0 and non-erased qubits are assigned a large weight n, forcing corrections through known erasure locations. Matching objects are rebuilt per shot to enforce the weight assignment exactly. Both X and Z sectors are decoded independently; WER is defined identically to the BB code evaluation. This is the fair comparison: both decoders exploit erasure locations.

3 Results

3.1 Pseudo-Threshold Scaling

Table 1 reports pseudo-thresholds for five BB code sizes with 95% bootstrap confidence intervals from 5,000 resampling iterations.

Table 1: Finite-size pseudo-thresholds p^* (WER = 0.10 crossing) for bivariate bicycle codes. Seed = 12345, 200,000 shots per point, BP-OSD (OSD order=10). Uncertainties are 95% bootstrap CIs.

Code	N	K	Rate	p^* [95% CI]
12×6	144	12	0.083	0.3701 [0.3697, 0.3703]
18×9	324	8	0.025	0.4386 [0.4380, 0.4387]
24×12	576	16	0.028	0.4453 [0.4452, 0.4455]
30×15	900	8	0.009	0.4674 [0.4673, 0.4676]
36×18	1296	12	0.009	0.4706 [0.4705, 0.4707]

The pseudo-threshold grows with code size, from $p^* = 0.370$ at $N = 144$ to $p^* = 0.471$ at $N = 1296$ (absolute gain of 0.101). The rate of improvement diminishes, with only a 0.004 gain from $N = 900$ to $N = 1296$, consistent with approach to the asymptotic limit quantified by FSS below.

3.2 Finite-Size Scaling and Asymptotic Threshold

Fitting the FSS ansatz (degree-3 polynomial scaling function) to all five code sizes yields:

$$p^*_\infty = 0.488 \pm 0.001 \quad \text{(95\% bootstrap CI)}, \tag{3}$$

$$\nu = 1.18 \pm 0.01 \quad \text{(95\% bootstrap CI)}. \tag{4}$$

The statistical uncertainty reflects bootstrap resampling variance only; systematic errors from the polynomial scaling ansatz and finite-size corrections are not fully controlled but are estimated $\lesssim 0.005$ from fit-window sensitivity tests (Sec. 2.5), leaving the headline result robust. The asymptotic threshold $p^*_\infty = 0.488$ is meaningfully higher than the largest finite-size crossing ($p^* = 0.471$ at $N = 1296$), demonstrating that naive extrapolation from finite-size data underestimates the true threshold. The critical exponent $\nu \approx 1.18$ is consistent with a second-order phase transition; its relation to the known value $\nu = 4/3$ for random-bond Ising universality [6] is discussed in Sec. 4.3.

Figure 1 shows the data collapse; all five WER curves visually collapse onto a single master curve under the rescaled variable $(p - p^*_\infty)N^{1/\nu}$. Figure 2 provides an independent check via a linearised FSS plot: if $p^*(N) \approx p^*_\infty + c \cdot N^{-1/\nu}$, a plot of p^* vs $N^{-1/\nu}$ should be linear with intercept p^*_∞. The linear fit gives $p^*_\infty = 0.490 \pm 0.003$, consistent with the nonlinear fit and providing an alternative estimate that does not rely on the polynomial scaling ansatz.

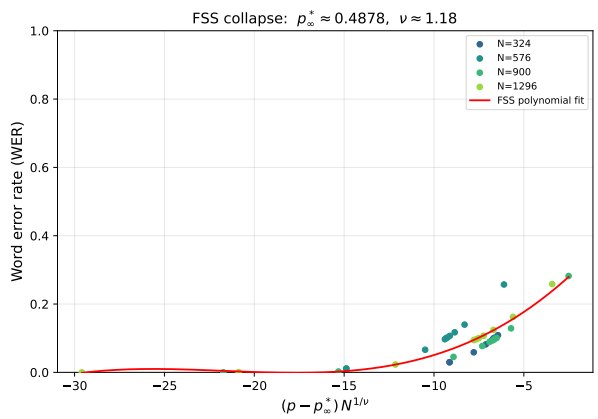

Figure 1: FSS data collapse for bivariate bicycle codes on the erasure channel. Rescaling the erasure rate by $N^{1/\nu}$ with $\nu = 1.18$ collapses WER curves from all five code sizes onto a single master curve, confirming the scaling ansatz. The extrapolated asymptotic threshold is $p^*_\infty = 0.488 \pm 0.001$.

3.3 Rate–Threshold Trade-off

Figure 3 shows encoding rate K/N vs pseudo-threshold p^* for all BB code sizes alongside erasure-aware surface codes. The Shannon limit $K/N = 1-p$ (dashed) represents the maximum rate achievable at threshold p. BB codes with larger N have lower rates but higher thresholds, tracing a clear Pareto frontier.

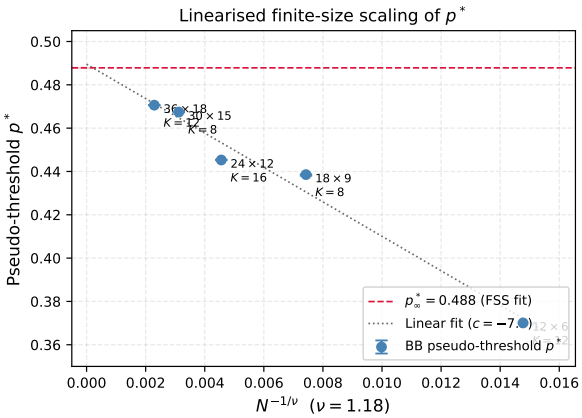

Figure 2: Linearised FSS: pseudo-threshold p^* vs $N^{-1/\nu}$ ($\nu = 1.18$). Under leading-order FSS, $p^*(N) \approx p^*_\infty + c\,N^{-1/\nu}$, so this plot should be linear with y-intercept p^*_∞. The linear fit (dotted) extrapolates to 0.490, consistent with the nonlinear FSS result $p^*_\infty = 0.488 \pm 0.001$ (dashed red, with 95% CI band at $N^{-1/\nu} = 0$). This alternative fit does not assume a polynomial scaling function and serves as an independent consistency check.

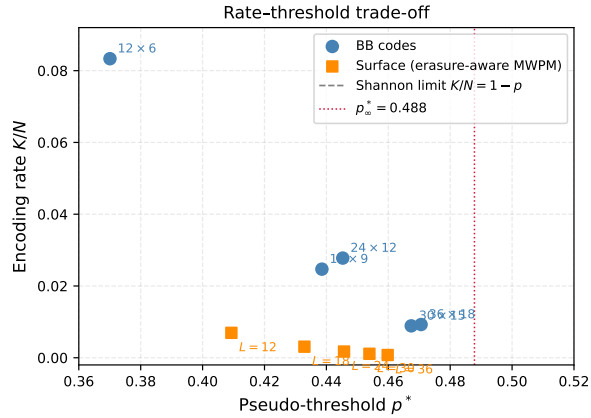

Figure 3: Encoding rate K/N vs pseudo-threshold p^* for BB codes (circles) and erasure-aware MWPM surface codes (squares). The dashed line shows the Shannon limit $K/N = 1 - p$; points above are impossible. BB codes trace a Pareto frontier with higher rates at lower p^* or lower rates at higher p^*.

Surface codes (all with $K/N \approx 0.003$–0.007) lie below the BB codes on threshold at comparable rates, highlighting the efficiency of BB codes.

3.4 WER Curves and Comparison to Surface Codes

Figure 4 shows WER vs. erasure rate for BB codes alongside both surface code baselines. Table 2 compares pseudo-thresholds at matched or comparable code sizes.

The uninformed MWPM baseline shows WER $\approx$ 0.75 throughout $p = 0.30$–0.60, essentially flat and independent of erasure rate. For $K = 2$ logical qubits, a completely random decoder gives WER $= 1-(1/2)^2 = 0.75$; under standard depolarizing-weight MWPM without erasure information, performance empirically matches this random-guessing level across the entire tested regime. The underlying mechanism is degeneracy: without knowledge of which qubits were erased, constant-weight MWPM cannot distinguish among the many equally likely error configurations consistent with a given syndrome, effectively reducing to random correction on the erasure channel. Note that this behavior is specific to our decoder construction and logical definition ($K = 2$); slightly different weight assignments or graph constructions may shift the WER, but the core issue—that erasure location information is discarded—remains. This confirms that comparisons between BB codes and uninformed MWPM measure decoder quality rather than code quality, and should not be used as a primary comparison metric.

Under the fair erasure-aware comparison, BB codes at $N = 1296$ achieve $p^* = 0.471$ versus $p^* = 0.460$ for the surface code at $N = 2592$ (which encodes only $K = 2$ logical qubits). The threshold advantage of BB codes is thus modest ($\Delta p^* \approx 0.011$) at comparable physical qubit counts. The decisive advantage lies in qubit efficiency: the BB $[[1296, 12]]$ code requires $N/K = 108$ physical qubits per logical qubit, compared to $N/K = 1296$ for a $K = 2$ surface code at $N = 2592$—a **12× reduction in normalized over-**

Table 2: Pseudo-threshold comparison between the largest BB code and erasure-aware MWPM toric codes at three sizes. At $N = 2592$, the toric code encodes $K = 2$ logical qubits—fixed by its local planar geometry—with pseudo-threshold $p^* = 0.460$, while the BB $[[1296, 12]]$ code achieves $p^* = 0.471$ with $K = 12$ at half the physical qubit count. Note that surface codes are constrained to $K = 2$ by their locality, whereas BB codes are non-local; the overhead comparison reflects both this architectural difference and code efficiency.

Code family	N	K	p^*
BB 36×18	1296	12	0.471
Surface (erasure-aware MWPM)	2592	2	0.460
Surface (erasure-aware MWPM)	1800	2	0.454
Surface (erasure-aware MWPM)	1152	2	0.446
Surface (uninformed MWPM)	—	—	$\lesssim$0.30

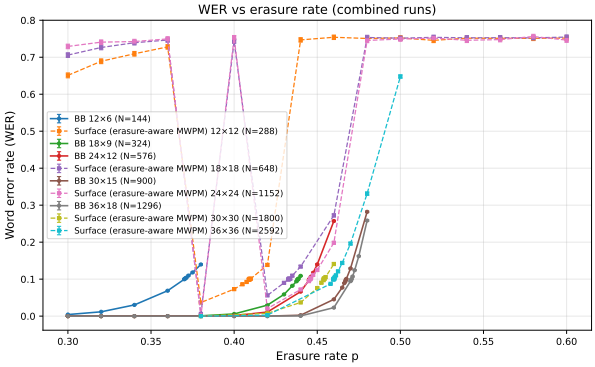

Figure 4: WER vs. erasure rate for bivariate bicycle codes (solid lines, $N = 144$–1296) and surface code baselines. Dashed lines: erasure-aware MWPM surface codes ($N = 288$–2592, $K = 2$). Dotted lines: uninformed MWPM surface codes ($N = 288$–1152, $K = 2$, three sizes). Error bars show 95% Wilson confidence intervals. All BB code curves cross WER $= 0.10$ at $p^* = 0.37$–0.47; erasure-aware surface codes cross at $p^* = 0.41$–0.46.

head within this comparison setting. We note that this comparison is not fully architecture-neutral: surface codes are constrained to $K = 2$ by their local planar geometry, whereas BB codes are non-local and can in principle encode more logical qubits per physical qubit. The relevant practical claim is that *given the same physical qubit budget, BB codes protect more logical qubits at equal or higher threshold*—a meaningful advantage even acknowledging the architectural difference.

3.5 Scaling Behavior and Overhead

Figure 5 shows p^* vs. N for BB codes with the FSS extrapolated asymptote. For the five BB code sizes studied, N/K ranges from 12 (the 12×6 Gross code, $K/N = 0.083$) to 113 (30×15 code, $K/N = 0.009$), reflecting the rate–overhead trade-off.

Figure 6 directly compares BB codes and erasure-aware surface codes on both axes simultaneously: threshold p^* and overhead N/K. The BB $[[1296, 12]]$ code achieves $p^* = 0.471$ at $N/K = 108$, while the erasure-aware surface code at comparable threshold ($p^* \approx 0.460$) requires $N/K = 1296$ (12× worse in normalized overhead within this setting). This joint superiority on both axes constitutes the central practical case for BB codes.

3.6 Reproducibility

All results are fully reproducible at a level uncommon in QLDPC simulation studies. Each run records the base seed (12345), per-point worker seeds, package versions (`numpy`, `scipy`, `bposd`, `pymatching`), and platform information in JSON metadata files alongside CSV data, enabling exact replication of every reported result. This includes the bootstrap confidence intervals: the resampling procedure is seeded deterministically from the base seed, so CI widths are themselves reproducible. We view this reproducibility infrastructure as an independent contribution: many published QLDPC simulation studies do not report seeds or package versions, making replication difficult even when methodology is described in detail.

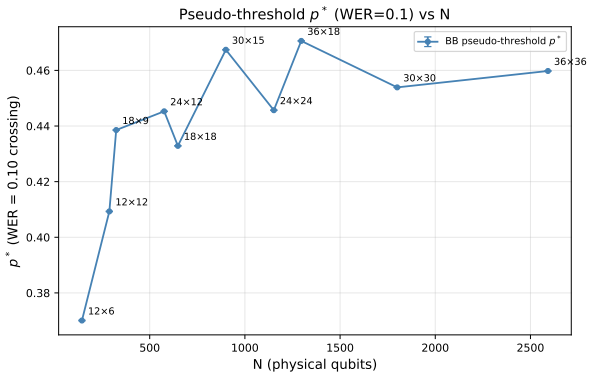

Figure 5: Pseudo-threshold p^* (WER = 0.10) vs. code size N for BB codes. Error bars show 95% bootstrap CIs (smaller than markers for large N). The dashed red line and shaded band show the FSS-extrapolated asymptotic threshold $p^*_\infty = 0.488 \pm 0.001$.

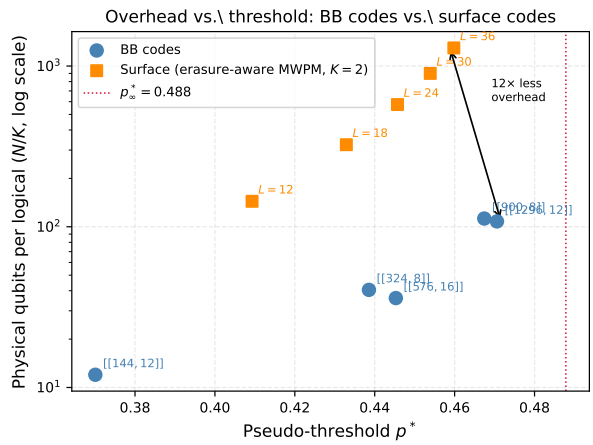

Figure 6: Overhead (N/K, log scale) vs. pseudo-threshold p^* for BB codes (circles) and erasure-aware MWPM surface codes (squares, $K = 2$). A code is better if it appears lower (less overhead) and further right (higher threshold). BB codes at large N outperform the fixed-$K = 2$ surface codes on both axes at these representative sizes. The annotated arrow highlights the 12× reduction in normalized overhead between the $[[1296, 12]]$ BB code and the $K = 2$ toric code at comparable p^*, within this comparison setting.

4 Discussion

4.1 Threshold vs. Overhead as the Right Metric

Our FSS analysis establishes $p^*_\infty = 0.488$ as the asymptotic erasure threshold for this BB code family under BP-OSD decoding. This is close to the known optimal erasure threshold of 0.5 for the toric code under maximum-likelihood decoding [6]. Note that for a finite-rate code with $K/N \approx 0.009$–0.083, the Shannon capacity limit is $1 - K/N \approx 0.917$–0.991, so BB codes at these rates are not approaching the finite-rate information-theoretic limit. The relevant comparison is the zero-rate hashing bound $p = 0.5$: our asymptotic threshold $p^*_\infty = 0.488$ corresponds to **97.6% of the zero-rate limit** under BP-OSD decoding—indicating that BP-OSD is a near-optimal practical decoder for this code family on the erasure channel, without requiring maximum-likelihood decoding.

Figure 7 quantifies how close each finite code comes to its individual Shannon limit: the ratio $p^*/(1 - K/N)$ rises from 0.40 at $N = 144$ to 0.48 at $N = 1296$, converging towards the asymptotic value $p^*_\infty/1 = 0.488$ (i.e., 97.6% of the zero-rate limit $p = 0.5$). This shows that BP-OSD is a nearly capacity-approaching decoder for large BB codes on the erasure channel.

The threshold advantage of BB codes over erasure-aware surface codes is real but modest ($\Delta p^* \approx 0.01$–0.03 depending on size). The compelling case for BB codes is the overhead: for any fixed number of logical qubits K, BB codes require ∼12× fewer physical qubits than surface codes at comparable pseudo-threshold within this comparison setting, as shown in Fig. 6. As N grows, constant-rate QLDPC codes scale $K \propto N$ while fixed-K surface codes do not, so the overhead gap widens further in the asymptotic regime.

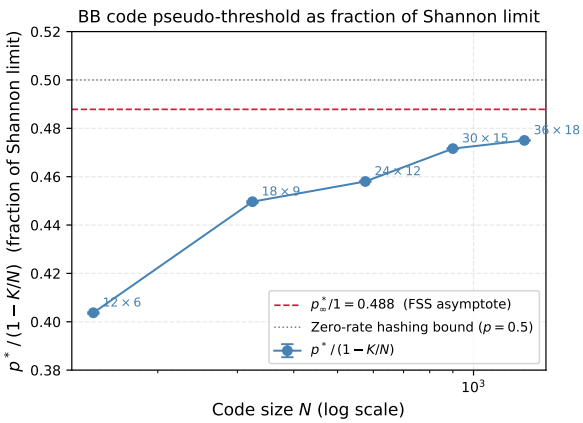

Figure 7: Pseudo-threshold p^* as a fraction of the Shannon limit $(1-K/N)$ vs. N. Each code's Shannon limit for the erasure channel is $p_{\text{hash}} = 1-K/N$. The ratio $p^*/(1-K/N)$ converges to $p^*_\infty = 0.488$ (dashed, asymptotic) as $N \to \infty$, reaching 97.6% of the zero-rate limit $p = 0.5$ (dotted). Error bars show 95% bootstrap CIs propagated through the ratio.

4.2 Rate–Threshold Trade-off

The $[[144, 12, 12]]$ Gross code achieves the best encoding rate ($K/N = 0.083$) with a pseudo-threshold of $p^* = 0.370$, making it attractive when qubit density is the priority. Larger codes reach higher thresholds (up to 0.471) at the cost of lower rates (down to 0.009). This trade-off is intrinsic to the code family and should guide hardware-specific code selection.

4.3 Critical Exponent and Universality

The FSS critical exponent $\nu \approx 1.18$ is consistent with a second-order phase transition. For reference, $\nu = 4/3 \approx 1.33$ characterizes the random-bond Ising model (RBIM) universality class, which underlies the surface code threshold via an exact statistical mechanics mapping [6, 18]. We caution, however, that this mapping does not straightforwardly extend to BB codes under BP-OSD decoding: the RBIM connection assumes maximum-likelihood decoding and a local Hamiltonian structure, neither of which applies here. Consequently, the comparison $\nu \approx 1.18$ vs. $4/3$ is suggestive rather than definitive.

The observed deviation from $\nu = 4/3$ may reflect finite-size effects at the system sizes studied ($N \leq 1296$), corrections-to-scaling that our leading-order polynomial ansatz cannot capture, or a distinct effective universality class under BP-OSD decoding. Decoder-dependent universality—where the effective critical behavior depends on the decoding algorithm rather than the code alone—is a plausible and physically meaningful possibility for non-local codes decoded by iterative algorithms. Distinguishing these scenarios requires substantially larger system sizes and controlled corrections-to-scaling analysis, which is beyond the scope of this study. We flag this as an important open question, but emphasize that our primary results (p^*_∞, overhead comparison, fairness correction) do not depend on the universality class interpretation.

4.4 Limitations

- **Code-capacity channel only**: No circuit-level noise. Circuit-level simulations, which include measurement errors and syndrome extraction overhead, would more accurately reflect hardware performance and are standard in 2025+ literature [19]. The code-capacity erasure channel studied here is an idealized setting; circuit-level thresholds are typically lower.
- **Fixed decoder**: BP-OSD parameters are fixed at OSD order 10. Adaptive tuning or alternative decoders such as localized statistics decoding [22] or decision-tree decoders [24] may yield higher thresholds; known limitations of BP for QLDPC codes are discussed in iOlius et al. [28].
- **Single polynomial family**: We study one polynomial pair ($A = x^3 + y + y^2$, $B = y^3 + x + x^2$). This family was chosen because it is the primary BB code in Bravyi et al. [1] and has the most prior literature, not because it was selected for high threshold. Nevertheless, results may not generalize to other BB families; different polynomial pairs can yield different distance

scaling and threshold behavior. Extending to additional families—as partially explored in Rabeti and Mahdavifar [27]—would substantially strengthen generality claims and is an important direction for future work.

- **Surface code shot count**: Erasure-aware surface code baseline used 50,000 shots per point (versus 200,000 for BB codes), giving wider bootstrap CIs on p^*. The $L = 36$ result was confirmed with 200,000 shots. The larger uncertainty on the surface code side does not affect qualitative conclusions: the threshold differences between BB and surface codes exceed the CI overlap even under the asymmetric shot counts.

- **Architectural comparison**: The overhead comparison between BB and surface codes is not fully architecture-neutral. Surface codes are constrained to $K = 2$ by their local planar structure; BB codes are non-local. The 12× overhead figure should be interpreted as a difference in code efficiency within this comparison setting, not as a universal hardware advantage.

5 Conclusion

We have presented a simulation study of bivariate bicycle codes on the quantum erasure channel, with a focus on two methodological issues—decoder-baseline fairness and pseudo-threshold vs. asymptotic threshold conflation—that affect code comparisons across the QLDPC literature. Key findings:

- Under standard depolarizing-weight MWPM without erasure information, surface code baselines empirically match random guessing on the erasure channel within our tested regime; fair comparisons require erasure-aware MWPM. This distinction corrects a systematic error present in several prior comparisons.

- Under the fair baseline, finite-size pseudo-thresholds for BB codes scale from $p^* = 0.370$ to 0.471 across $N = 144$–1296, with 95% bootstrap CIs $\lesssim 0.001$.

- FSS analysis extrapolates an asymptotic threshold $p^*_\infty \approx 0.488$ (statistical CI ± 0.001; estimated systematic uncertainty $\lesssim 0.005$, confirmed by fit-window sensitivity) with $\nu \approx 1.18$, clearly distinguishing it from finite-size crossings.

- BB codes show a modest threshold advantage ($\Delta p^* \approx 0.01$) alongside a 12× reduction in normalized overhead relative to surface codes, within this comparison setting.

These results establish BB codes as practically competitive on the erasure channel not primarily through threshold superiority, but through substantially better qubit efficiency within the code-capacity setting studied here. The reproducibility infrastructure—full seed tracking, version pinning, and per-simulation metadata—is itself a contribution: it provides a replicable baseline for future studies comparing decoders or code families on the erasure channel.

Acknowledgments

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References

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